

Agent-to-Agent (A2A) Protocol

Open standard for interoperable agent communication across frameworks and stacks

What is A2A?

A2A is an open standard that defines how AI agents can discover each other, exchange information, and delegate tasks across different frameworks and platforms.

Discovery

Agent Cards via `/.well-known/agent-card.json`

Communication

Message-based, streaming via SSE

Tasks

Long-running processes supported

Security

OAuth2, Bearer/Basic auth

Key Features

-  **Cross-platform interoperability**
Works with different frameworks and languages
-  **Enterprise readiness**
Multi-tenancy, signed Agent Cards
-  **Web-aligned architecture**

Benefits



Reduced Integration

No custom glue code needed



Team Collaboration

Decoupled teams & stacks



Scalability

Easy to add/modify agents



Maintainability

Focused testing & debugging

Microsoft Integration

Agent Framework

Microsoft.Agents.AI.Hosting.A2A

A2A Client

A2AAgent/A2AClient

A2A Communication Flow

Microsoft Agent Framework A2A — Use Cases

Real-world applications of Agent-to-Agent protocol in enterprise scenarios

Cross-Team Compliance

Procurement agent hands off to partner compliance agent via A2A without stack rewrites. Ensures policy adherence across organizational boundaries.

Enterprise

Security

✓ Active

Customer Support Triage

Triage agent delegates to billing/network specialists using A2A endpoints. Intelligent routing based on query analysis.

Support

AI

🕒 24/7

Connected Analytics

Orchestrator queries research, retrieval, summarizer agents across vendors. Aggregates results for comprehensive insights.

Analytics

RAG

✓ Active

Legacy Integration

Wrap legacy service as an A2A agent. Partners discover via Agent Card and call securely without API changes.

Integration

Legacy

🔄 Sync

Multi-Cloud Interop

Expose your agent; consume third-party A2A agents in the same workflow. Seamless cross-cloud communication.

Cloud

Hybrid

✓ Active

Workflow Automation

Automate complex business processes with A2A agents handling different stages of the workflow.

Automation

Workflow

🤖 Auto

 Implementation Flow

 Discovery →  Connection →  Communication →  Validation

`A2ACardResolver` → `GetAIAgentAsync()` → `RunAsync()`

Multi-Agent Orchestration

Building orchestrator agent with planner + executor + validator sub-agents

Agent Roles & Responsibilities



Planner

Decomposes goals into steps; selects agents/tools



Executor

Performs tasks, runs tools, executes actions



Validator

Checks outputs vs acceptance criteria

Workflow Diagram



Implementation Guidance

Sequential

Use for step-by-step processing with clear dependencies

Concurrent

Use for parallel processing and independent analysis

Group Chat

Use for collaborative problem-solving

Code Example

C#

```
var workflow = new AgentWorkflowBuilder ()
    . WithPlanner (plannerAgent)
    . WithExecutor (executorAgent)
    . WithValidator (validatorAgent)
    . Build ();
```